

Scalability & Future Plan

Date	12 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static dataset for prediction	May not reflect latest agricultural conditions	High
2	No real-time weather API integration	Recommendations don't consider live weather	High
3	No user login or history tracking	Users cannot save or review previous predictions	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud with load balancing to support more users
2	Data Storage	Static CSV dataset	Use a cloud database (MySQL/PostgreSQL/Firebase)
3	Performance	Single ML model running locally	Optimize model and use faster cloud deployment
4	Security	Basic input validation	Add user authentication, HTTPS, and secure data storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Integrate real-time weather API	1–2 Months	More accurate crop recommendations
Phase 2	Add fertilizer recommendation system	2–3 Months	Improve crop productivity
Phase 3	Develop Android mobile application	3–5 Months	Better accessibility for farmers